

TASK 4 / Day 4

Data Visualization & Storytelling Report

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Project Overview

The Sample-Superstore data set is used in this project to examine the sales performance of products and determine the major financial losses through product sub-categories and discounts.

Main Question and Response

Question:

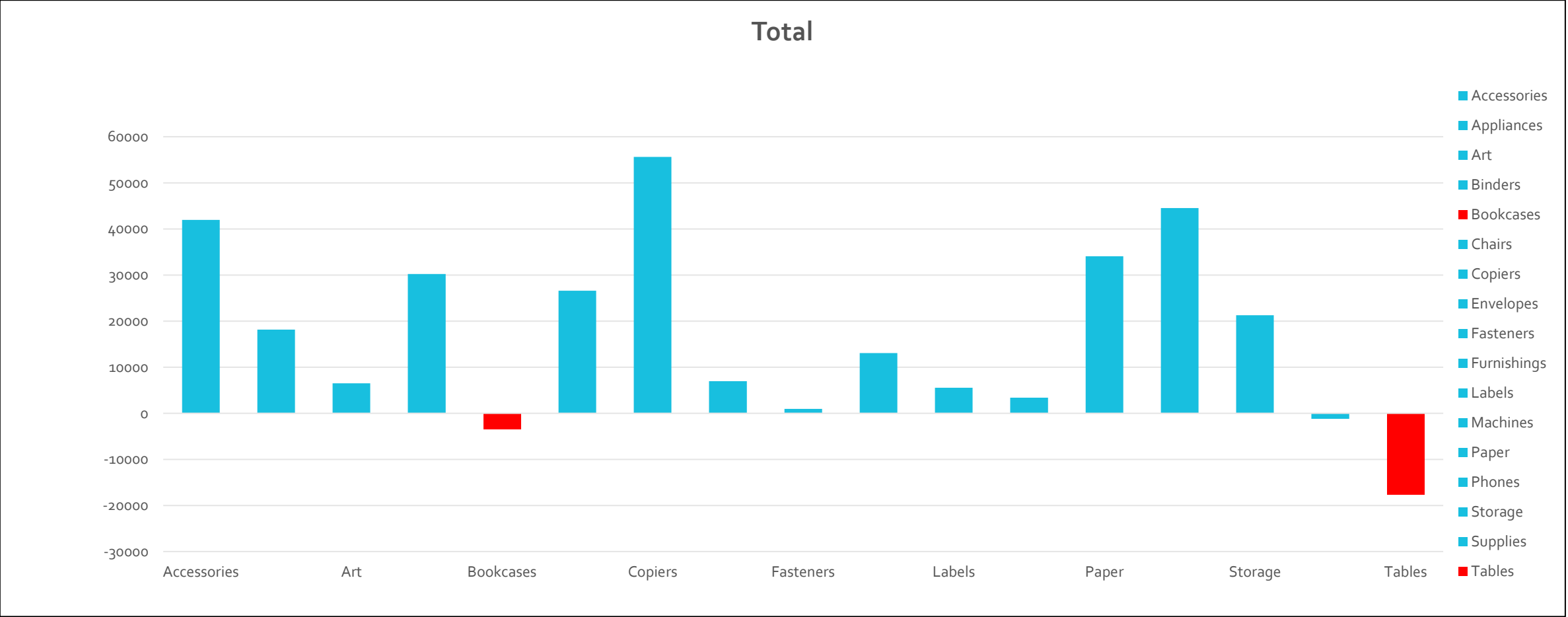
Why is it that some highly-sold product sub-categories incur massive financial losses despite high revenue from sales?

Answer:

Discounting beyond the 20 percent mark leads to a collapse of profit margins; hence, Tables (−\$17,725) and Bookcases (−\$3,473) are the main drivers of financial losses.

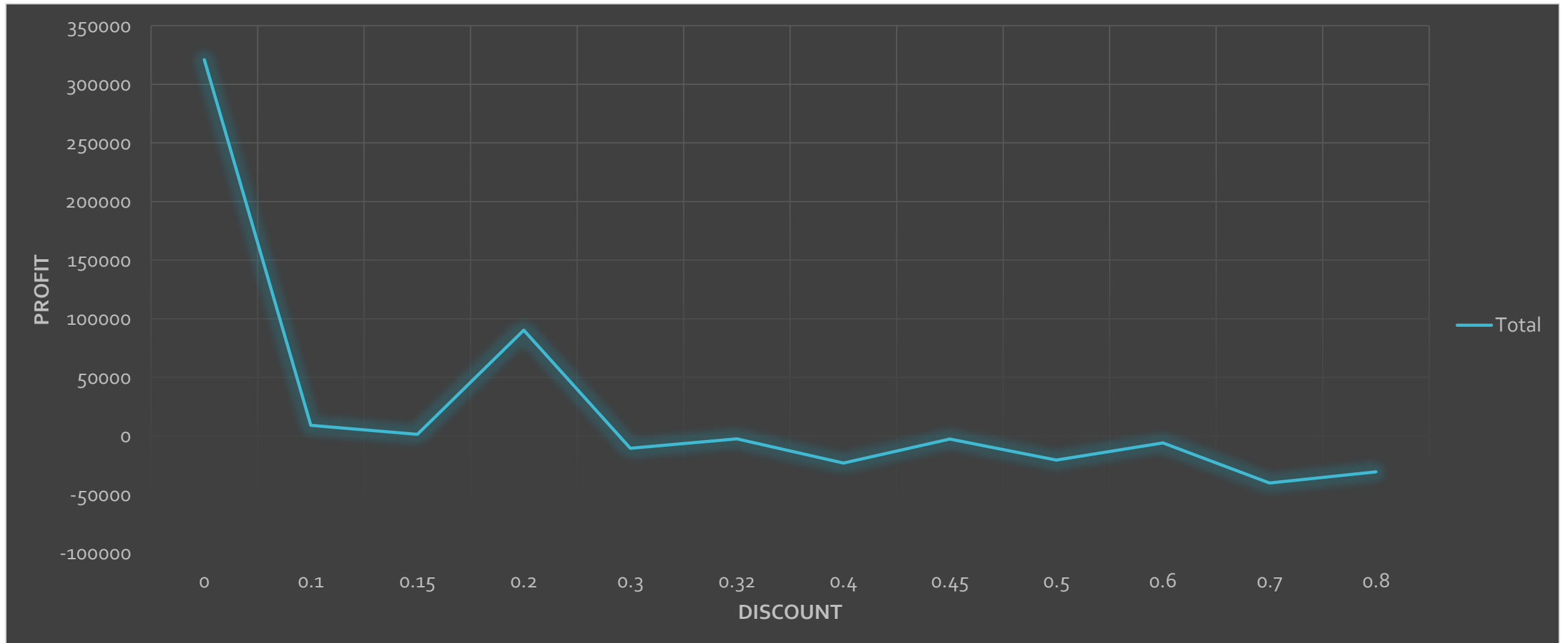
Chart 1-Profit and Loss based on Sub-Categories

Sabiya



Interpretation: While most sub-categories generate strong profits (led by Copiers at +\$55,618), Tables (-\$17,725) and Bookcases (-\$3,473) are highlighted in red as the primary drivers of financial loss.

Chart 2 - Discount vs Profit Analysis



Interpretation: Profit margins remain healthy at low discount levels but drop sharply below zero as soon as discounts cross the 20% (0.2) threshold, proving that aggressive discounting causes the financial losses.

Story Line

1. Context

The business demonstrates a strong core revenue model with robust profitability across 14 of its 17 product sub-categories. High-performing technology and office supply categories—such as Copiers (+\$55,618), Phones (+\$44,516), and Accessories (+\$41,937)—generate consistent, healthy margins that establish a solid operational foundation.

2. The Problem

Overall company profit margins are heavily undermined by severe, localized financial losses concentrated in the office furniture segment. Tables (-\$17,725) represents the single largest net loss driver across the entire store, followed by Bookcases (-\$3,473), creating a major drag on total net income despite generating substantial raw sales revenue.

3. Evidence

The analysis of the factors influencing price behavior shows that loss of profits is directly associated with high discount pricing of products and not demand for the products. Discounts within the range of 0% to 20% produce stable positive profit margins. But once the discount percentage exceeds 20% (0.2), then margins become highly negative, leading to higher sales producing even greater losses.

4. Recommendations

In order to avoid further loss to the company and recover its gross margin, it becomes necessary to implement restrictions on pricing discounts.

It is essential to set up the restriction on 15% discount limit on all products of furniture so that the discounts don't go into negative margin and prevent losses from Tables and Bookcases.